Max Hlavacek

Curriculum Vitæ
Current as of October 12, 2026

2339 Estella Laboratory, Pomona College
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Research Interests

Enumerative, geometric, and algebraic combinatorics, including lattice point enumeration problems for polyhedra.

Appointments

2023- **Pomona College**
Visiting Assistant Professor of Mathematics and Statistics

Education

2023 **University of California, Berkeley**
Ph.D., Mathematics
Advisor: Matthias Beck and Mark Haiman
Title: Ehrhart Theory of Combinatorially Defined Polytopes

2017 **Harvey Mudd College**
B.S., Mathematics

Publications

Articles

2026 **Ehrhart h^*-distributions**
B. Braun, M. Hlavacek, C. J. Meza, S. Morales, A. R. Vindas-Meléndez, arXiv Preprint

2025 **The poset of maximal tubings of the cycle graph is a lattice**
B. Adenbaum, E. Barnard, M. Hlavacek, B. Kagy, N. Lesevich, G. Nasr, K. Waddle, arXiv Preprint

2025 **Out of the Parking Lot and into the Forest: Parking Functions, Bond Lattices, and Unimodal Forests**
J. Brooks, S. Fishel, M. Hlavacek; S. Rubenfeld, B. C. Teves, *Discrete Mathematics*, Vol. 348 Iss. 12, pp. 1-16

2024 **Polyhedral geometry of refined q,t-Catalan numbers**
M. Beck, M. Hanada, M. Hlavacek, J. Lentfer, A. R. Vindas-Meléndez, K. Waddle, to appear in *Journal of Combinatorics*, pp. 1-27

2023 **Inequalities for f^*-vectors for lattice polytopes**
M. Beck, D. Deligeorgaki, M. Hlavacek, J. Valencia-Porras, *Advances in Geometry*, 24(2), pp.141-150.

2022 **Subdivisions of Shellable Complexes**
M. Hlavacek, L. Solus. *Journal of Combinatorial Theory, Series A* Vol. 186, pp.1-45

2018 **Random matrix ensembles with split limiting behavior**
P. Burkhardt-Guim, P. Cohen, J. Dewitt, M. Hlavacek, S.J. Miller, C. Sprunger, Y. N. Truong Vu, R. Van Peski, K. Yang, and an appendix joint with M. Fernandez and N. Sieger. *Random Matrices: Theory and Applications* Vol. 7 No. 3., pp. 1-32

2018 **Summand minimality and asymptotic convergence of generalized Zeckendorf decompositions**
K. Cordwell, M. Hlavacek, C. Huynh, S.J. Miller, C. Peterson, Y. N. Truong Vu. *Research in number theory*, Vol. 4, No. 43, pp. 1-27

2016 **A decomposition of parking functions by undesired spaces**
M. Dougherty, M. Bruce, M. Hlavacek, R. Kudo, and I. Nicolas. *Electronic journal of combinatorics*, Vol. 23, Iss. 3, pp. 1-22

Conference Proceedings

2023 **Inequalities for f^*-vectors for lattice polytopes**
M. Beck, D. Deligeorgaki, M. Hlavacek, J. Valencia-Porras, *Séminaire Lotharingien de Combinatoire* 89B.43 Article 43 (Proceedings of FPSAC 2023)

2022 **Subdivisions of Shellable Complexes**
M. Hlavacek, L. Solus. *Séminaire Lotharingien de Combinatoire* 85B, Article 29 (Proceedings of FPSAC 2021)

2019 **Cubical Dehn-Sommerville equations and self-reciprocal cubical complexes**
M. Hlavacek, *Proceedings of the 2018 Summer Workshop on Lattice Polytopes*, World Scientific.

Theses

2023 **Ehrhart Theory of Combinatorially Defined Polytopes**
M. Hlavacek, *University of California, Berkeley ProQuest Dissertations & Theses*, pp. 1-104

Presentations

Invited Talks

An Invitation to Ehrhart Theory
2026 Mathematical Pathways to an Excellent Future
2024 Practicum for Undergraduate Mathematicians in Combinatorics, IPAM
2023 Carleton College Math Colloquium
2023 Gettysburg College Math Colloquium
2022 Stanislaus State: A Speaker Series for the Quantitatively Curious

A Polyhedral View of q,t-Catalan numbers
2025 AMS Spring 2025 Eastern Sectional, Special Session on Recent Developments in Algebraic and Combinatorial Aspects of Polytopes
2025 Discrete Math Seminar, Arizona State University
2024 Claremont Colleges Mathematics Colloquium

f^*-vectors of lattice polytopes
2024 AMS Fall 2024 Western Sectional, Special Session on GEMS in Combinatorics, Riverside
2023 Claremont Colleges Algebra, Number Theory, and Combinatorics Seminar

Parking Functions, Bond Lattices, and Unimodal Forests
2023 JMM AWM Special Session on Women in Graph Theory
2022 UC Berkeley Combinatorics Seminar

Signed Poset Polytopes

2022 UC Davis Algebra and Discrete Math Seminar
2022 Algebra, Combinatorics, and Geometry Seminar SFSU
2022 JMM, AMS Session: Geom. and Top. Combinatorics
2021 Gems of Combinatorics
2021 Graduate Online Combinatorics Colloquium, Online
2020 Combinatorics Seminar, Otto von Guericke University Magdeburg
2019 Commutative Algebra and Lattice Polytopes, RIMS, Kyoto University

Subdivisions of Shellable Complexes

2023 Characteristic Polynomials of Hyperplane Arrangements and Ehrhart Polynomials of Convex Polytopes, RIMS, Kyoto University
2022 NY Combinatorics Seminar (online)
2022 University of Washington Combinatorics and Geometry Seminar
2022 Online Combinatorics Seminar, Washington University
2021 AMS Session: Geom. and Top. Comb. and their Applications, Online
2020 Combinatorics Seminar, Freie Universität Berlin
2019 Commutative Algebra and Lattice Polytopes, RIMS, Kyoto University

Cubical Dehn-Sommerville equations

2019 Lattice polytopes, with a view towards Geometry and Applications, Edinburgh
2019 UC Berkeley Combinatorics Seminar
2019 San Francisco State University AGC Seminar

Contributed Talks and Posters

A Polyhedral View of q,t-Catalan numbers

2026 Poster Presentation at the CIMPA School on Combinatorial Structures in Geometry, Graphs, and Symmetry, University of Costa Rica

Maximal tubings of the cycle graph

2025 MAA Socal and Nevada Section Meeting, CSU Fullerton

f^*-vectors of lattice polytopes

2025 Poster presentation at IPAM Workshop on Computational Interactions between Algebra, Combinatorics, and Discrete Geometry

Subdivisions of Shellable Complexes

2022 Poster presentation at Formal Power Series and Algebraic Combinatorics, online
2021 Poster presentation at an Algebraic Combinatorics Virtual Expedition, online

Signed Poset Polytopes

2021 Queer and Trans Mathematicians in Combinatorics Conference, Fields Institute

Cubical Dehn-Sommerville equations

2018 Summer Workshop on Lattice Polytopes, Osaka
2018 Poster presentation at Einstein Workshop on Geometric and Topological Combinatorics, Berlin

Teaching

Pomona College

Fa. 2026 Math 102: Differential Equations and Modeling
2024-2026 Math 103: Combinatorics
2023-2026 Math 30: Calculus 1
Fa. 2025 Math 101: Introduction to Real Analysis
2024-2025 Math 60: Linear Algebra

Pomona College Academy for Youth Success

Su. 2026 Math and problem solving for rising high school sophomores

UC Berkeley Lead Instructor

2022 Lead Instructor for Math 32: Precalculus
2019 Summer Instructor for Math 55: Discrete Mathematics

UC Berkeley Graduate Student Assistant

2023 GSI for Math 185: Complex Analysis
2022 Head GSI for Math 10B: Methods of Mathematics: Calculus, Statistics and Combinatorics
2021 GSI for Math 74: Transition to Upper Division Mathematics
2021 GSI for Math 10B: Methods of Mathematics: Calculus, Statistics and Combinatorics
2020 GSI for Math 10A: Methods of Mathematics: Calculus, Statistics and Combinatorics
2018-2019 GSI for Math 54: Linear Algebra and Differential Equations

Mentoring and Advising

Undergraduate Summer Research

2025 Summer Research at Pomona, Research Advisor for Anya Lin and Cris Ovalle,
Total order signed chain polytopes
2022 Summer @ ICERM: Computational Combinatorics, research TA under Susanna Fishel
for Josephine Brooks, Sophie Rubenfeld, and Bianca Carmelita Teves

Undergraduate Theses

2025-2026 Landen Isacson, *in progress*
2025-2026 Elizabeth Newman, *in progress*
2025-2026 Kai Sivasankar, *in progress*
2025-2026 Kellie Au, Triangle counting in social networks
2025-2026 Philip Belin, On Ehrhart positivity of order polytopes
2025-2026 Jake Sugawara, Re-Purposing math education: Practices and understanding to support students
2024-2025 Zoe Piccirillo, Agent-based modeling of climate policy: methods and applications
2024-2025 Clara Caspard, Graph optimization and related topics in algebraic geometry
2023-2024 Louis Burns, Ehrhart theory of some signed poset polytopes

Other

2026 CIMPA School on Combinatorial Structures in Geometry, Graphs, and Symmetry (teaching assistant)
2021 Combinatorial Polynomials at REACT Online Conference (teaching assistant)
2020 Directed Reading Program Mentor at UC Berkeley
2018 Counselor at Summer Mathematics Research Training, Texas AM University
2017 Counselor at PROMYS, Boston University

Workshops and Professional Development

Activities Organized

2025 Co-organized Pizza and Progress Summer Research Seminar, Pomona College
2025 Co-organized AMS Special Session on MRC Algebraic Combinatorics at JMM 2025, Seattle

2021	Co-organized 41st Bay Area Discrete Math Day, Online
2021	Co-organized Talking about Combinatorial Objects Student Seminar, UC Berkeley

Activities Attended

2026	CIMPA School on Combinatorial Structures in Geometry, Graphs, and Symmetry at the University of Costa Rica
2025	IPAM Workshop on Computational Interactions between Algebra, Combinatorics, and Discrete Geometry
2024	Mathematics Research Community in Combinatorics: Algebraic Combinatorics, Java Center NY
2023	Project NExT at 2023 MathFest, Tampa
2023	GEMs of Combinatorics at AIM, San Jose
2022	Ehrhart polynomials: inequalities and extremal constructions at AIM, San Jose
2020	Combinatorial Coworkspace - A session in algebraic and geometric combinatorics, Kleinwalsertal, Austria
2019	AG+ Research Retreat, Dollnsee, Germany
2019	Facets of Complexity Annual Workshop, Dollnsee, Germany
2018	Summer Workshop on Lattice Polytopes, Osaka, Japan

Awards and Certificates

2025	Summer Undergraduate Research Program Grant, Pomona College
2024	Green'23 Project NExT Fellow
2020	Certificate of Remote Instruction, UC Berkeley GRI Fellows Program
2019	Outstanding GSI Award, UC Berkeley
2017	Chancellor's Fellowship, UC Berkeley
2017	Greever Mathematical Research Prize, Harvey Mudd College Mathematics Department

Service

2023-2025	Organized events for Professional Development Network for Gender Minorities in Math and Computing at the Claremont Colleges
2016-2025	Refereed papers for Forum of Mathematics, Sigma,Proceedings of the 2018 Summer Workshop on Lattice Polytopes at Osaka University, and Fibonacci Quarterly
2024	Invited speaker at Practicum for Undergraduate Mathematicians in Combinatorics
2022	Invited Speaker at OURFA^{2}M^2 '22
2021-2023	Core Member of Lavender Phoenix
2020-2023	Healing Justice Committee Member at Lavender Phoenix
2019 - 2020	Equity and Inclusion Committee member at UC Berkeley Math Department
2018-2021	Unbounded Representation (URep) Member at UC Berkeley
2017-2018	Volunteer with Youth Support Program at Willard Middle School
2017-2023	Member of Noetherian Ring